COMSATS University Islamabad (CUI)

Project Proposal

for

OVULITE

Version 1.0

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Bachelor of Science in Data Science (2023-2027)

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Project Category: (Select all the major domains of proposed project**)**

☐ **A-**Desktop Application/Information System	☐ **B-**Web Application/Web Application based Information System	
☐ **C-**Problem Solving and Artificial Intelligence	☐ **D-**Simulation and Modeling	☐ **E-**Smartphone Application
☐ **F-**Smartphone Game	☐ **G-**Networks	☐ **H-**Image Processing
☑ Other (specify category): IVF		

Abstract

The rapid expansion of IVF and embryo transfer (ET) in Pakistan's cattle industry has exposed critical challenges in reproductive decision-making, where technicians and embryologists still rely on subjective visual assessment and manual judgment. This results in inconsistent pregnancy outcomes, misuse of embryos, and significant economic losses for farms adopting assisted reproductive technologies. The proposed system aims to resolve these challenges by enabling objective, data-driven prediction of pregnancy success, early identification of abortion risk, and improved embryo-recipient matching to optimize reproductive efficiency. A major grey area in current practice is the absence of an integrated AI platform that analyzes existing embryology, donor, recipient, technician, and hormonal data to support accurate on-ground decisions. **Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle** seeks to overcome this gap by building machine-learning models for pregnancy prediction, abortion risk assessment, and AI-based embryo grading using image-metadata fusion. The system further aims to provide reproductive analytics, donor–sire performance insights, and technician effectiveness reports, allowing farms to improve outcomes using the data they already collect. Ultimately, this project will enhance overall reproductive success, reduce embryo wastage, and support Pakistan's transition toward precision livestock reproduction.

1. Introduction

The purpose of this project proposal is to outline the design and development of **Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle**, a data-centric solution aimed at improving the accuracy and efficiency of assisted reproductive processes in livestock. With the increasing adoption of IVF (In-Vitro Fertilization) and ET (Embryo Transfer) technologies in Pakistan, farms and embryology labs are generating large amounts of valuable reproductive data that remain underutilized due to the absence of intelligent analytical tools. Current decision-making practices rely heavily on manual assessment of embryo quality, recipient suitability, hormonal status, and technician skill, which often results in inconsistent reproductive outcomes. Variability in CL size, blood flow, progesterone levels, embryo day, and recipient physiology further complicates the process, making success prediction difficult without systematic data-driven support. Additionally, farms lack structured analytics to evaluate the performance of donors, sires, embryos, recipients, media batches, or technicians over time, leading to repeated inefficiencies and increased operational costs. Technological advancements in machine learning, image analysis, and predictive modeling offer opportunities to transform this domain through automated embryo grading, pregnancy prediction, and abortion risk assessment. Ovulite aims to integrate these modern AI-based capabilities into a unified system that leverages existing field and laboratory data to support better reproductive decisions. The system intends to provide embryologists, veterinarians, and farm managers with actionable insights derived from CL measurements, blood flow scores, P4 levels, embryo characteristics, and historical ET outcomes. By bridging the gap between practical livestock operations and advanced computational methods, Ovulite aspires to introduce precision reproduction into Pakistan's cattle industry. This proposal details the problem, objectives, scope, methodologies, and expected contribution of the system in enhancing the reliability of cattle IVF and ET programs.

2. Problem Statement

The current reproductive management practices in Pakistan's cattle IVF and embryo transfer industry rely heavily on subjective assessment, intuition, and manual interpretation of biological indicators. Embryologists and field veterinarians often evaluate embryo grade, recipient suitability, CL size, blood flow, and hormonal profiles based on visual judgment rather than standardized, data-driven criteria. This leads to inconsistent pregnancy outcomes, higher rates of early embryonic loss, and inefficient use of expensive donor-derived embryos. Furthermore, IVF labs lack analytical tools to correlate embryology parameters—such as embryo day, media type, oocyte yield, and technician involvement—with pregnancy success, making it difficult to identify performance bottlenecks. Recipient selection remains a trial-and-error process because farms do not have systems that combine CL size, progesterone levels, blood flow, and physiological history to calculate suitability or predict embryo acceptance. The absence of automated monitoring and prediction mechanisms contributes to

avoidable embryo wastage and financial losses, especially when frozen embryos or high-value donor genetics are used. Moreover, farms have no structured way of tracking abortion risk using early hormonal patterns, blood-flow changes, or falling P4 trends, resulting in late detection of failing pregnancies. This lack of predictive intelligence prevents timely interventions and increases overall reproductive failure rates.

Despite farms collecting substantial data during OPU, IVF, ET, and pregnancy checks, there is no integrated platform that analyzes these datapoints holistically to guide technicians and managers. As a result, opportunities for optimizing donor–sire pairings, recipient selection, embryo grading consistency, and technician performance remain unmet. The development of an AI-powered system that leverages existing data to provide accurate predictions, actionable insights, and reproductive intelligence is therefore essential to improving success rates, reducing losses, and enabling precision livestock reproduction.

3. Problem Solution/Objectives of the Proposed System

The proposed system, **Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle**, is designed to introduce objective, data-driven decision-making into Pakistan's cattle IVF and embryo transfer workflow. By leveraging the dataset already collected during OPU, IVF, ET, and pregnancy monitoring, the system replaces subjective judgment with quantifiable, reproducible, and AI-supported assessments. The core of the solution is a machine-learning-based pregnancy prediction model that uses CL size, blood flow, progesterone level, embryo day, embryo grade, donor-sire metadata, and technician history to estimate the likelihood of successful conception. This prediction empowers veterinarians and embryologists to prioritize high-probability transfers and avoid wasting valuable embryos on unsuitable recipients. Ovulite also includes an abortion risk prediction module that analyzes early hormonal patterns, CL regression, weight changes, and blood-flow signals to alert users about high-risk pregnancies.

The system further addresses embryo-quality inconsistencies through an AI-backed embryo grading model using image–metadata fusion that enhances grading accuracy and reduces inter-technician variability. An embryo–recipient matching engine optimizes the allocation of embryos by recommending top three recipients based on physiological, hormonal, and historical ET factors. In addition, Ovulite performs donor and sire performance analytics to help farms identify top genetic combinations and plan more effective breeding strategies. Technician performance evaluation enables IVF labs to monitor operator-level variation and improve SOPs through feedback loops. The system also analyzes protocol effectiveness, allowing users to compare outcomes of fresh versus frozen embryos, ET timing, or synchronization protocols.

By centralizing all reproductive data into a single unified platform, Ovulite transforms scattered farm records into actionable intelligence. It introduces predictive analytics, early warning alerts, and performance dashboards that significantly enhance reproductive efficiency. In doing so, the system aims to reduce embryo losses, improve conception rates, support timely interventions, and provide a repeatable methodology for high-quality ET operations. Ultimately, Ovulite enables

precision livestock reproduction through modern artificial intelligence and systematic reproductive analytics.

3.1 Objectives

BO-1: Develop a machine-learning model to predict embryo pregnancy success using CL size, blood flow, P4, embryo metadata, and technician input.
BO-2: Build an abortion risk alert system to identify pregnancies prone to early loss based on hormonal and physiological indicators.
BO-3: Implement an AI-based embryo grading system using image–metadata fusion to improve grading accuracy and reduce technician subjectivity.
BO-4: Design an embryo–recipient matching engine to recommend the most suitable recipients for each embryo.
BO-5: Analyze synchronization protocol effectiveness (e.g., Conventional vs Sexed embryos, CIDR, PGF2α, GnRH) to support evidence-based IVF and ET decision-making.
BO-6: Integrate all data sources into a unified reproductive intelligence dashboard for farms and IVF labs.
BO-7: Reduce embryo wastage and improve overall pregnancy success through actionable, data-driven insights.

4. Related System Analysis/Literature Review

Several digital and AI-enabled livestock solutions exist globally, but none provide an integrated embryo-level reproductive intelligence system specifically designed around Pakistan's cattle IVF and ET workflows. **EmGenysis** is an embryo monitoring and management platform offering digital embryo records and donor tracking; however, it lacks AI-based pregnancy prediction, abortion risk alerts, and recipient suitability analysis, making it insufficient for data-driven ET decisions. **Folio3 AI Livestock Management** provides herd monitoring, disease alerts, and vision-based animal identification, but does not support IVF-related analytics, embryo grading, or ET success modeling. **EmbryoPlus**, a commercial bovine reproduction service, offers advanced IVF services and donor management but does not provide farm-level predictive models or AI-driven embryo–recipient matching based on physiological and hormonal data. **Taurus Herd Health** focuses on herd fertility monitoring and insemination tracking, yet it does not incorporate embryo image analysis, donor–sire matrix optimization, or technician performance analytics.

These limitations highlight the global gap in integrated, AI-supported embryo-level reproductive intelligence. Ovulite addresses this gap by merging embryo-quality assessment, pregnancy prediction, abortion-risk modeling, recipient suitability scoring, and donor/sire analytics into one unified reproductive intelligence system powered by machine learning and image fusion models.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Folio3 AI Livestock Management	Focuses on herd health and monitoring, not IVF/ET-specific reproductive analytics or embryo-quality evaluation.	Ovulite integrates embryo grading (image + metadata fusion), donor–recipient analytics, and ET-focused predictive models.
EmbryoPlus	Provides IVF services but lacks predictive intelligence, in-lab embryo viability scoring, and recipient suitability modeling.	Ovulite adds AI embryo grading, CL/P4-based suitability scoring, technician analytics, and donor–sire sweet spot analysis.
Taurus Herd Health	Tracks fertility cycles but does not handle embryo images, IVF workflows, ET success prediction, or donor–sire matrix.	Ovulite extends functionality with advanced embryo analytics, ML prediction models, and reproductive intelligence dashboards.
EmGenysis	Offers embryo records but no AI pregnancy prediction, abortion risk assessment, or embryo–recipient matching.	Ovulite provides ML-based pregnancy prediction, abortion alerts, and AI-driven embryo matching to improve ET success rates.

5. Vision Statement

For cattle farms, IVF laboratories, embryologists, and livestock reproduction specialists Who require accurate, data-driven predictions to improve embryo transfer success and reduce reproductive losses The *Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle* Is an integrated reproductive analytics and decision-support platform That uses machine learning, embryo image analysis, hormonal indicators, and historical ET data to predict pregnancy outcomes, assess abortion risk, grade embryos with AI, and recommend optimal embryo–recipient matches Unlike existing herd management tools, manual embryo grading, or subjective technician assessments Our product offers a unified embryo-to-pregnancy intelligence system that delivers precise predictions, real-time reproductive alerts, donor–sire performance insights, and operator analytics — all derived from the farm's existing IVF and ET datasets.

6. Scope

The scope of Ovulite – AI-Driven Embryo Success Prediction and Reproductive Intelligence System for Cattle encompasses the development of a unified decision-support platform that analyzes

data generated throughout the IVF–ET–pregnancy cycle. The system will process donor, sire, embryo, recipient, technician, hormonal, and pregnancy outcome data already collected by farms and IVF laboratories to deliver actionable reproductive intelligence. Key functionalities include machine-learning-based pregnancy success prediction, abortion risk assessment, AI-powered embryo grading using image–metadata fusion, and an optimized embryo–recipient matching assistant. Ovulite will generate reproductive analytics, such as donor performance ranking, sire compatibility matrices, protocol effectiveness, embryo day/stage success trends, and technician performance insights to support evidence-based management decisions. The system will visualize CL size, blood flow, progesterone levels, embryo quality, and ET outcomes to help identify sweet spots and failure patterns in existing reproductive workflows.

Additionally, Ovulite will incorporate a dashboard for monitoring reproductive KPIs, including OPU-to-pregnancy funnel analytics, embryo batch health, and embryo wastage patterns. The platform will provide a structured interface for embryologists to enter embryo metadata and images, and for veterinarians to update recipient profiles and pregnancy outcomes. The solution does not include real-time sensor integration, genomic analysis, drone imaging, environmental prediction, or automated physiological monitoring, as these are beyond the dataset and resources available in the current project. It also excludes financial modules unrelated to reproduction, such as full herd management, milk yield tracking, or feedlot operations.

The project will deliver a functional prototype containing core predictive models, visual dashboards, and user interfaces required for practical use by IVF labs and farm managers. Ovulite will be deployed as a web-based application and may include a supportive chatbot for explanation and guidance. By defining clear functional boundaries, the project ensures feasibility within available data, academic timeline, and computational resources while still offering high-impact AI-driven reproductive intelligence for cattle.

7. Modules

The proposed system consists of several core modules that collectively deliver AI-driven reproductive intelligence for cattle. Each module is designed to transform raw IVF–ET–pregnancy data into meaningful predictions, alerts, and analytics. The system is divided into functional components that cover embryo assessment, pregnancy prediction, abortion risk identification, donor/sire analytics, protocol effectiveness, and operator performance evaluation. The following modules and their features define the precise boundaries of the Ovulite system.

7.1 Module 1: Pregnancy Success Prediction Engine

FE-1: Predict pregnancy outcome using CL size, blood flow score, P4 level, embryo metadata, technician, and protocol.

FE-2: Display prediction probability as percentage and RAG (Red–Amber–Green) score.

FE-3: Provide SHAP-based explanatory factors for each prediction.
FE-4: Compare prediction against similar historical cases for validation.

7.2 Module 2: Log- in System

FE-1: Secure user login ensures only authorized individuals can access the system.
FE-2: Access is controlled through role-based permissions, allowing users to view and manage features according to their assigned roles.
FE-3: Active session management maintains user security and ensures smooth, uninterrupted system usage.
FE-4: Secure logout functionality protects user data by safely ending sessions when access is no longer needed.

7.3 Module 3: AI Embryo Grading (Image + Metadata Fusion)

FE-1: Upload embryo images (Day 6/7/8).
FE-2: Extract image-based features using a CNN (e.g., ResNet).
FE-3: Combine image features with metadata (day, donor, media, embryo stage).
FE-4: Predict embryo grade (1/2/3) and viability score.
FE-5: Display visual heatmaps showing embryo regions contributing to grade.

7.4 Module 4: LAB QC indicator

What it does: Finds "something is off" in lab performance without waiting months.
Inputs (you already have)
Media used
Technician
Embryo day/stage/grade
Oocyte count
Blast/embryo yield (if present)
Pregnancy outcome (if present)
Date / batch ID (even if you approximate by week or session)
How it works (different from predictive models)
Unsupervised anomaly detection (Isolation Forest / One-Class SVM) + control charts
Detects "this week/media lot/technician shift is deviating from baseline"
Detects model drift: embryo grades/pregnancy rates shifting over time
Outputs
"Media X shows 18% lower outcomes than baseline"
"Technician B outcomes dropped significantly in last 20 cycles"
"This batch is abnormal — investigate incubator, media prep, vitrification"
✅Why it's separate: It's not predicting pregnancy for a single ET; it's monitoring lab process health like an audit system.
✅Why IVF labs love it: It's directly SOP + audit value.

7.5 Module 5: Protocol Effectiveness Analysis

FE-1: Compare outcomes for conventional vs sexed embryos.
FE-2: Evaluate synchronization protocols (CIDR, PGF2α, GnRH).
FE-3: Analyze technician × protocol interactions.
FE-4: Provide protocol success dashboards with pregnancy % and abortion %.

7.6 Module 6: Reproductive Intelligence Dashboard

FE-1: Visualize CL-size sweet spots, blood-flow comparisons, and P4 curves.
FE-2: Show embryo day/stage vs pregnancy rates.
FE-3: Display IVF funnel (Oocyte → Embryo → ET → Pregnancy).
FE-4: Track embryo batch-level issues (media lots, operator effects).
FE-5: Provide overall reproductive KPIs in a unified dashboard.

8. System Limitations/Constraints

LI-1: The system is limited to the dataset currently available, which includes donor, sire, embryo, recipient, hormonal, technician, and pregnancy outcome information; it cannot train models requiring genomics, environmental data, feed intake, or sensor-based physiological measurements.
LI-2: Ovulite does not integrate real-time IoT devices, automated ultrasound imaging, or continuous hormonal monitoring, as such data is not accessible within the scope of this project.
LI-3: The AI embryo grading module depends on the quality and quantity of embryo images provided; inconsistent lighting, focus, or image resolution may affect grading accuracy.
LI-4: Machine learning predictions will be constrained by the size and diversity of the dataset, and may not generalize perfectly across all breeds, technicians, or farms without further data expansion.
LI-5: The system is designed for cattle IVF and ET workflows and does not extend to full herd management such as feeding schedules, milk yield tracking, disease monitoring, or financial reporting.
LI-6: Ovulite is a decision-support system and does not replace professional veterinary or embryology expertise; its predictions must be interpreted by qualified personnel.
LI-7: The prototype will be a web-based application subject to internet availability, bandwidth constraints, and device compatibility limitations on rural farms with weak connectivity.

9. Data Gathering Approach

Data for this project is being gathered from the IVF laboratory at **DayZee Farms**, which has been actively performing OPU, IVF, and ET procedures for over two years. The farm maintains detailed logs of donors, embryos, recipients, hormone values, technician information, and pregnancy

outcomes, which serve as the primary dataset for Ovulite's predictive models. Additional insights are collected through brief interviews and discussions with embryologists and veterinarians to understand workflow requirements and decision-making patterns. This ensures that the system is built on authentic, real-world reproductive data and operational practices.

10. Tools and Technologies

Table 2: Tools and Technologies for Proposed Project

	Tools and Technologies	Version	Rationale
Tools and Technologies	**Frontend Development**		
	Next.js	13 (Latest)	A React-based framework used for building server-side rendered applications. Provides SEO optimization and faster page load times, ideal for the web app.
	HTML5	Latest	Core markup language used to structure content on the web. Essential for building web pages with semantic elements.
	CSS3	Latest	Styling language used to create the layout, design, and responsive structure of web pages, ensuring compatibility across different devices and screen sizes.
	JavaScript	ES6 (Latest)	A scripting language used to add interactivity to websites. Handles frontend logic, DOM manipulation, and API requests.
	Backend Development		
	Node.js	18 (LTS)	JavaScript runtime environment for server-side scripting. Allows for building full-stack applications using JavaScript.
	Express.js	4.17.1	A web framework for Node.js used for building RESTful APIs and handling routing and middleware in backend development.
	Flask	2.0.2	A Python-based micro-framework used to build lightweight REST APIs. Works well for integrating machine learning models into the backend.
	Machine Learning		
	TensorFlow	2.8	An open-source machine learning library used for building neural networks, such as for pregnancy prediction and embryo grading.
	Keras	2.8	A high-level API for building and training deep learning models. Simplifies model development, particularly for image-based tasks.
	Scikit-learn	1.0.2	Python library is used for traditional machine learning algorithms such as classification, regression, and clustering.
	PyTorch	1.11	Deep learning framework for training complex neural networks. Primarily used for image analysis tasks like embryo grading.
	Pandas	1.4.2	Python library is used for data manipulation and analysis. Essential for preprocessing and cleaning data before feeding it into machine learning models.
	NumPy	1.22.4	Core library for numerical computing. Handles arrays and matrices, essential for working with data in machine learning models.
	Database		
	MySQL	8.0	An open-source relational database management system. Perfect for storing structured data such as embryos, donors, recipients, and pregnancy outcomes.

	MongoDB	5.0	A NoSQL database used for storing unstructured or semi-structured data. Good for handling flexible schemas, but might not be necessary for this project.
	Image Processing		
	OpenCV	4.5.5	Open-source computer vision library. Used for analyzing and processing embryo images for feature extraction, helping with tasks like grading embryo quality.
	Testing and debugging		
	Postman	9.1	A platform for testing APIs. Allows easy creation and testing of HTTP requests and responses. Essential for ensuring API functionality during development.
	UI/UX Design Tools		
	Figma	Free Plan	A collaborative design tool used for creating wireframes, prototypes, and interactive mockups. Ideal for frontend design and client feedback.

11. Project Stakeholders and Roles

Table 3 Project Stakeholders for Proposed Project

Stakeholders	**Responsibilities**
Project Sponsor: DayZee Farms	Provides project funding, access to farm data, equipment, and technical resources. Owns all rights to the final product.
Stakeholder	Provides academic supervision, evaluation, and project guidance within the FYP framework.
Farm Manager	Shares operational data, validates system outputs, and provides feedback for real-world usability.
Veterinarian	Supplies embryo transfer (ET), pregnancy, and abortion records; ensures accuracy of health-related data.
Student Developers: Eman Malik Inshal Zafar	Designs, develops, and implements the data collection and analysis system; manages database and dashboards.
Project Supervisor: Mam Nusrat Shaheen	Guides technical development, reviews documentation, and ensures academic and professional standards.
Final Year Project Committee	Evaluates project progress, implementation, and final outcomes for academic grading and compliance.

12. Module based Work Division

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Module / Feature
Eman Malik	SP23-BDS-012	Module 1: Embryo Success Prediction – Development of machine learning models to predict pregnancy success, integration of prediction logic into the backend.
		Module 2: Data Visualization & Analytics – Creation of visual dashboards to display predictive analytics, embryo data trends, and pregnancy success insights.
		Module 3: User Interface & Experience Design – Designing and testing the frontend layout, optimizing the user experience, and ensuring ease of navigation for farm staff and medical professionals.
Inshal Zafar	SP23-BDS-021	Module 4: Data Management & Database Integration – Design and implementation of the database schema, ensuring smooth data storage for embryos, recipients, and pregnancy outcomes.
		Module 5: Machine Learning & Model Integration – Training machine learning models for embryo grading, integrating models into the system, and fine-tuning prediction accuracy.
		Module 6: System Testing & Deployment – Conducting testing phases, including unit testing and debugging, followed by deploying the application on the cloud platform.

13. WBS and GANTT CHART

13.1 Work Based Structure(WBS)

The Work Breakdown Structure (WBS) for the **Ovulite** project is organized into three hierarchical levels to ensure clear scope definition, effective planning, and controlled execution.

Level 1 – Major Phases

Level 1 represents the major phases of the software development lifecycle. These phases define the overall structure of the project and group related activities at a high level.

The major phases of the Ovulite project are:

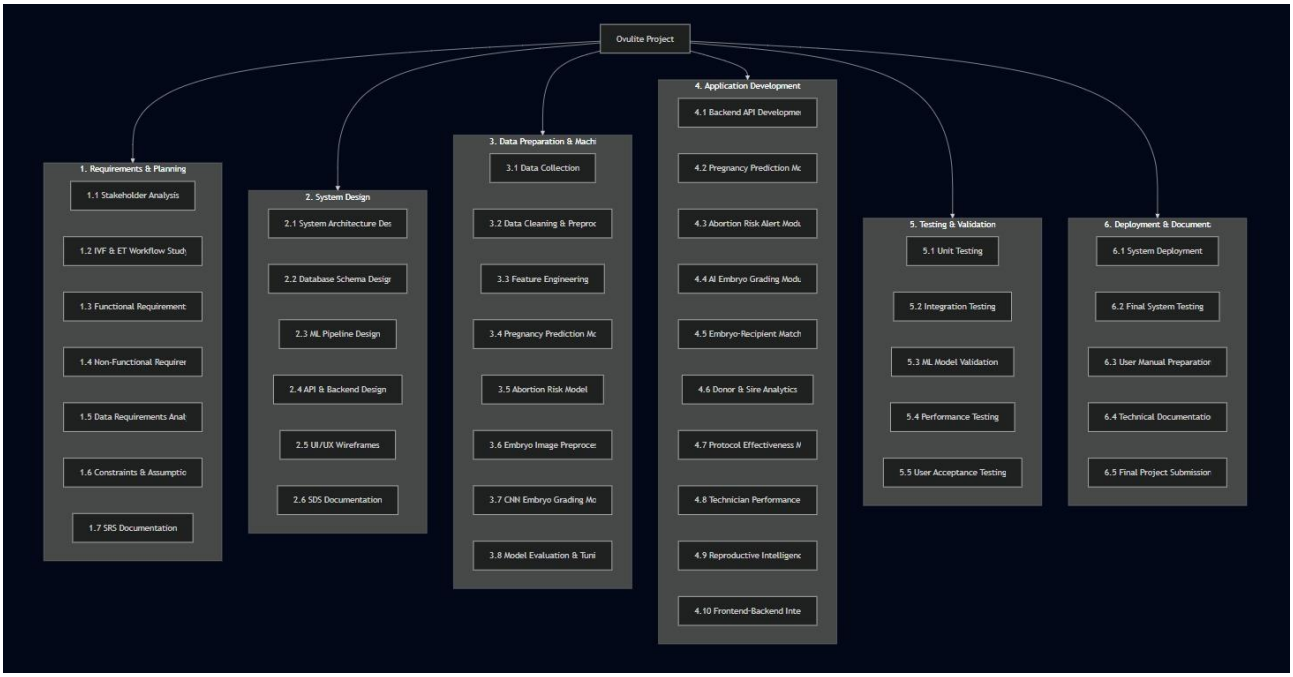
- Requirements & Planning
- System Design
- Data Preparation & Machine Learning
- Application Development
- Testing & Validation
- Deployment & Documentation

Level 2 – Sub-Activities

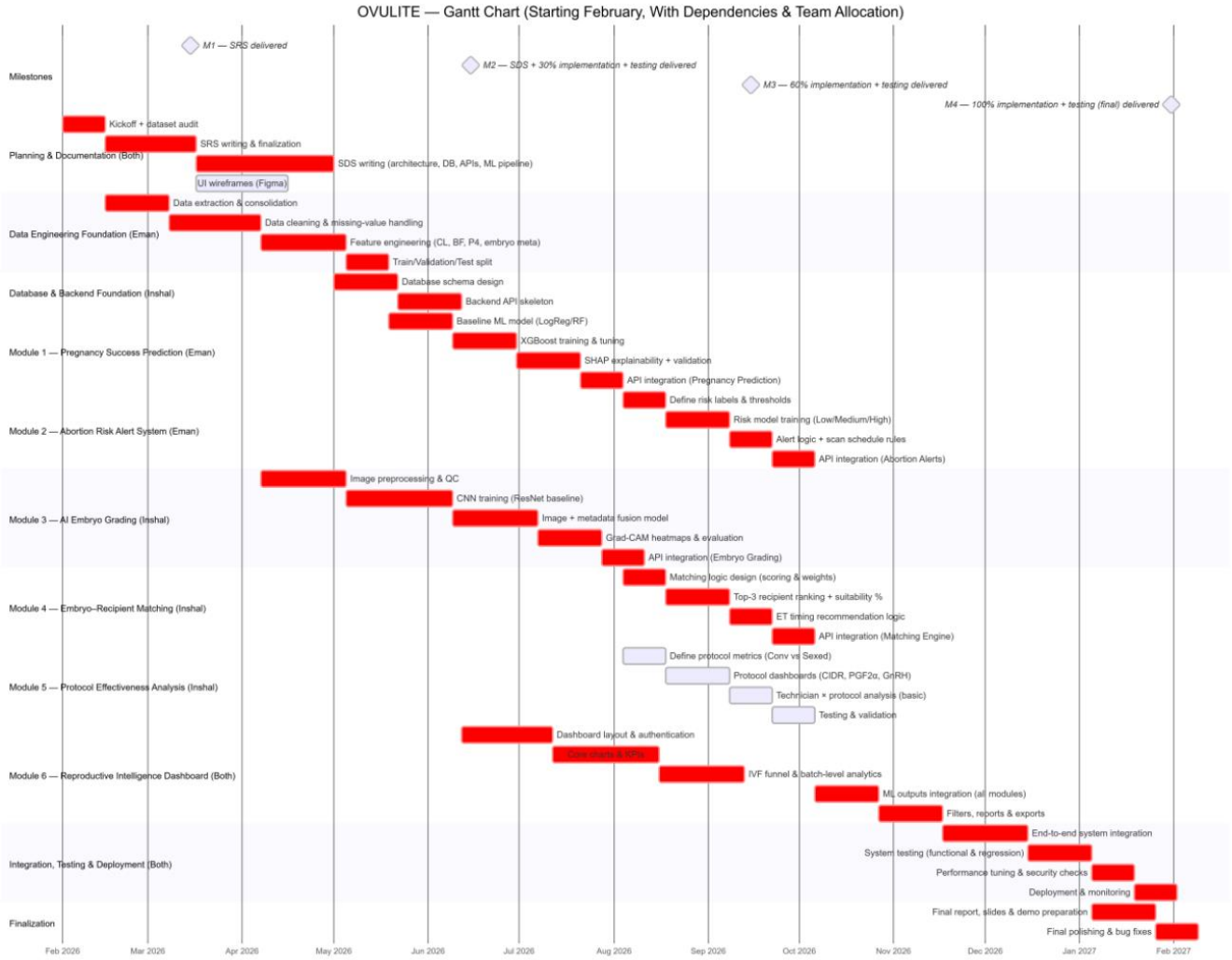
Level 2 decomposes each major phase into logical sub-activities that describe the key functional areas within that phase. These activities represent significant tasks required to complete each phase.

Examples include:

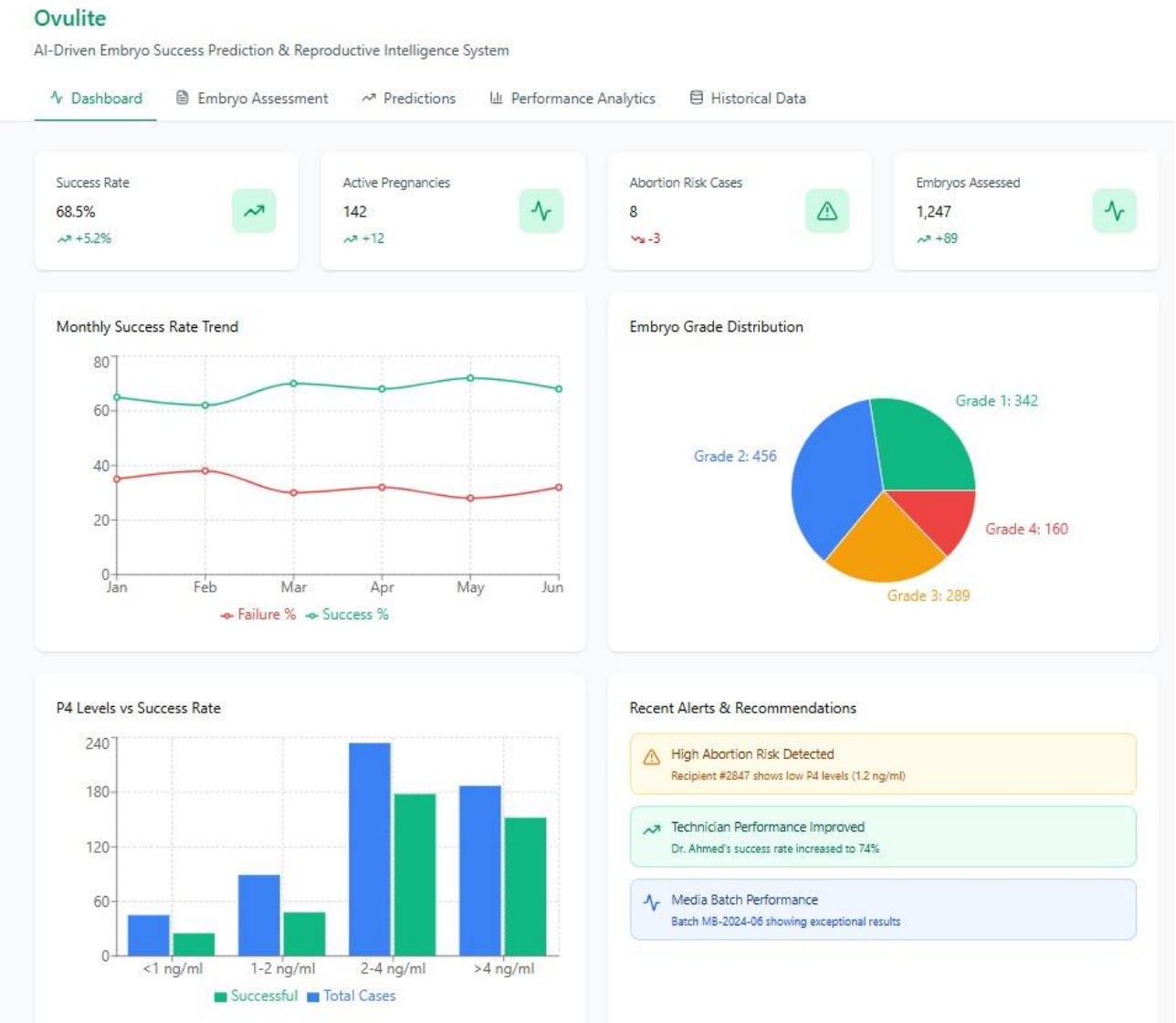
- Requirements analysis and documentation within Requirements & Planning
- Architecture, database, and UI design within System Design
- Data preprocessing and model development within Data Preparation & Machine Learning
- Module-wise implementation within Application Development

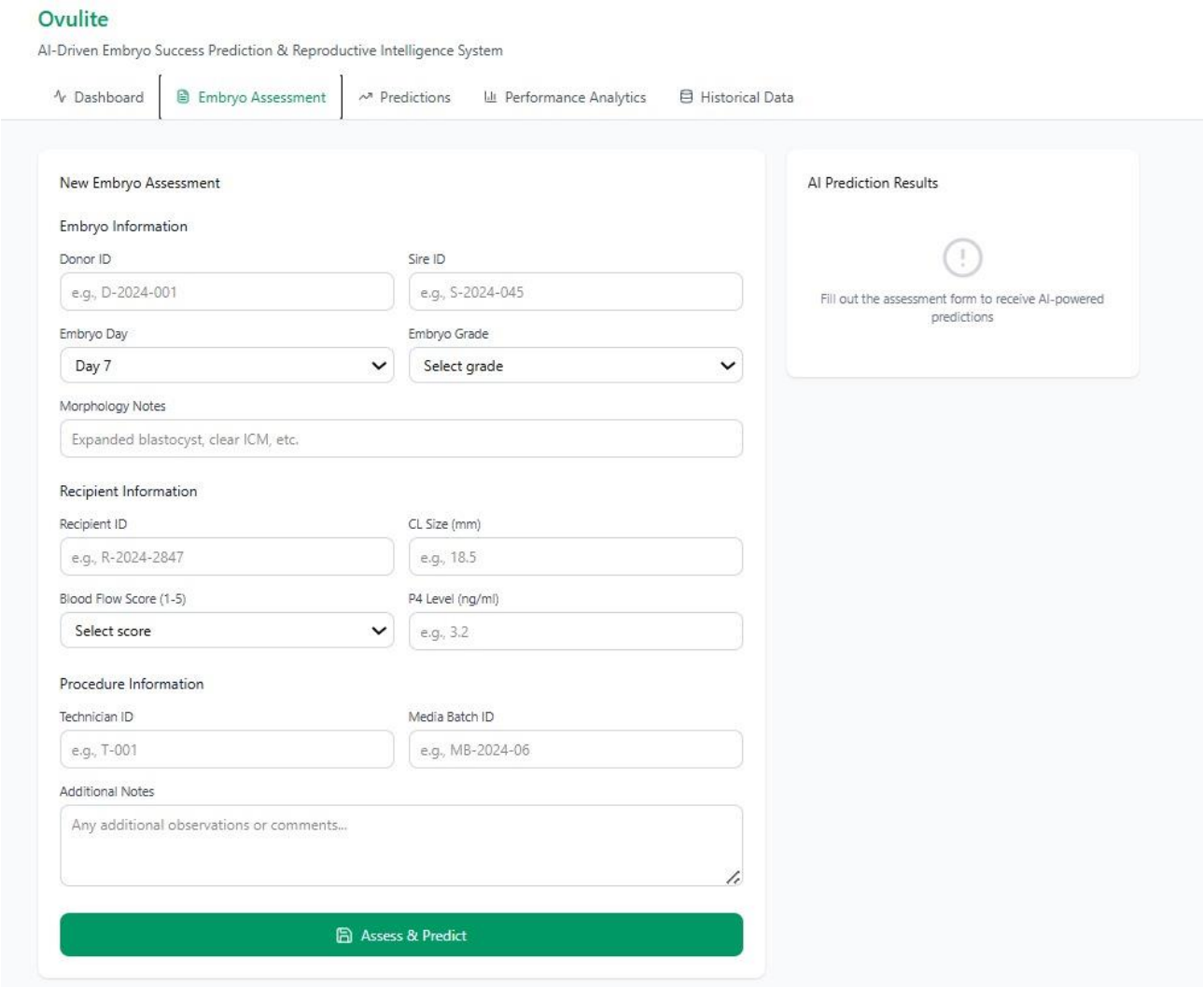


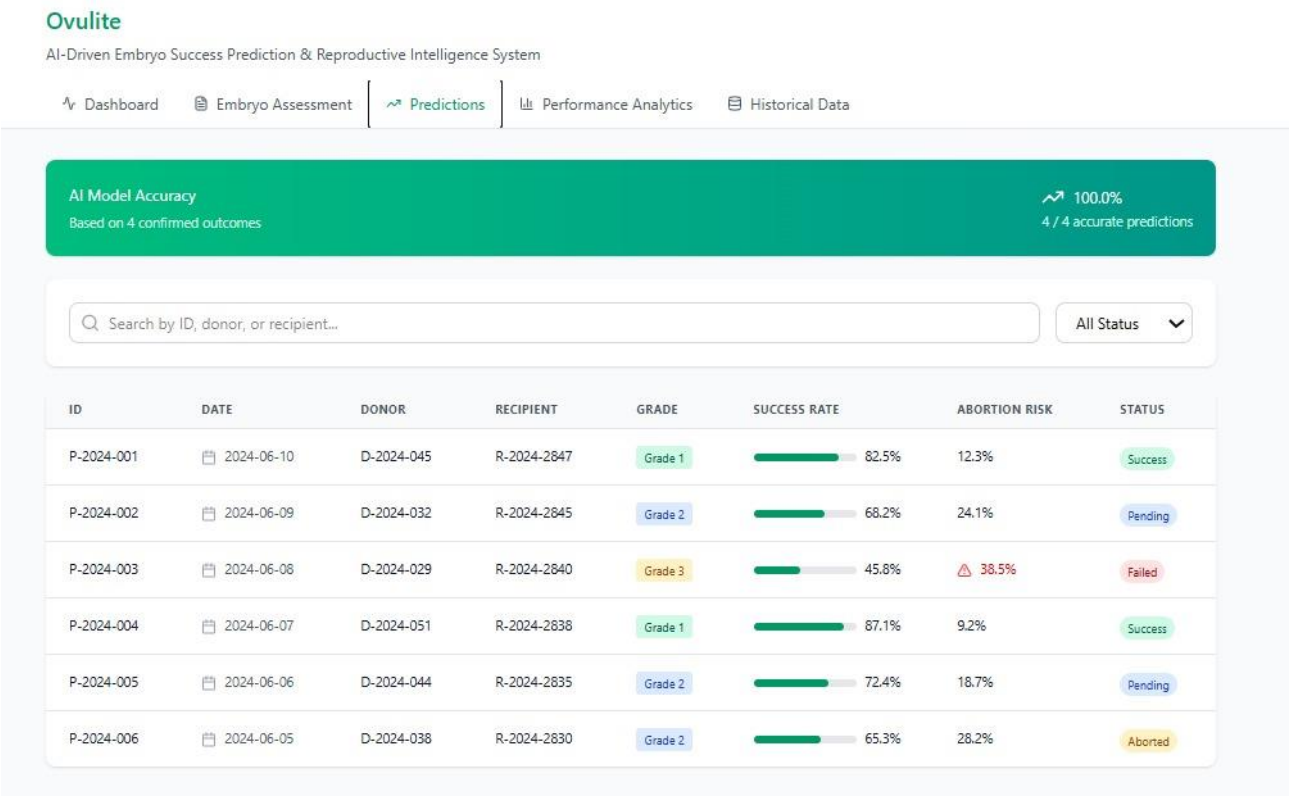
13.2 GANTT Chart

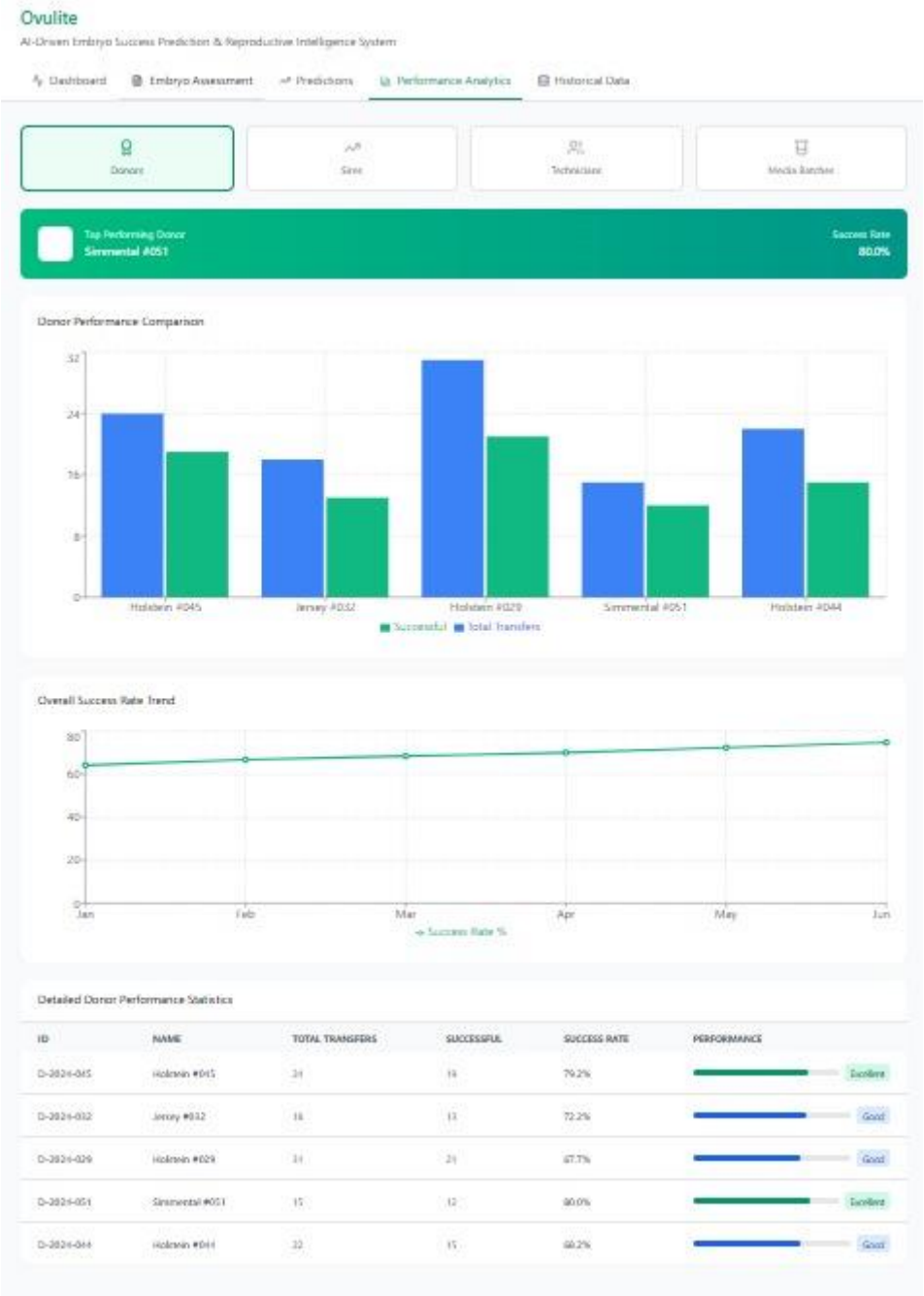


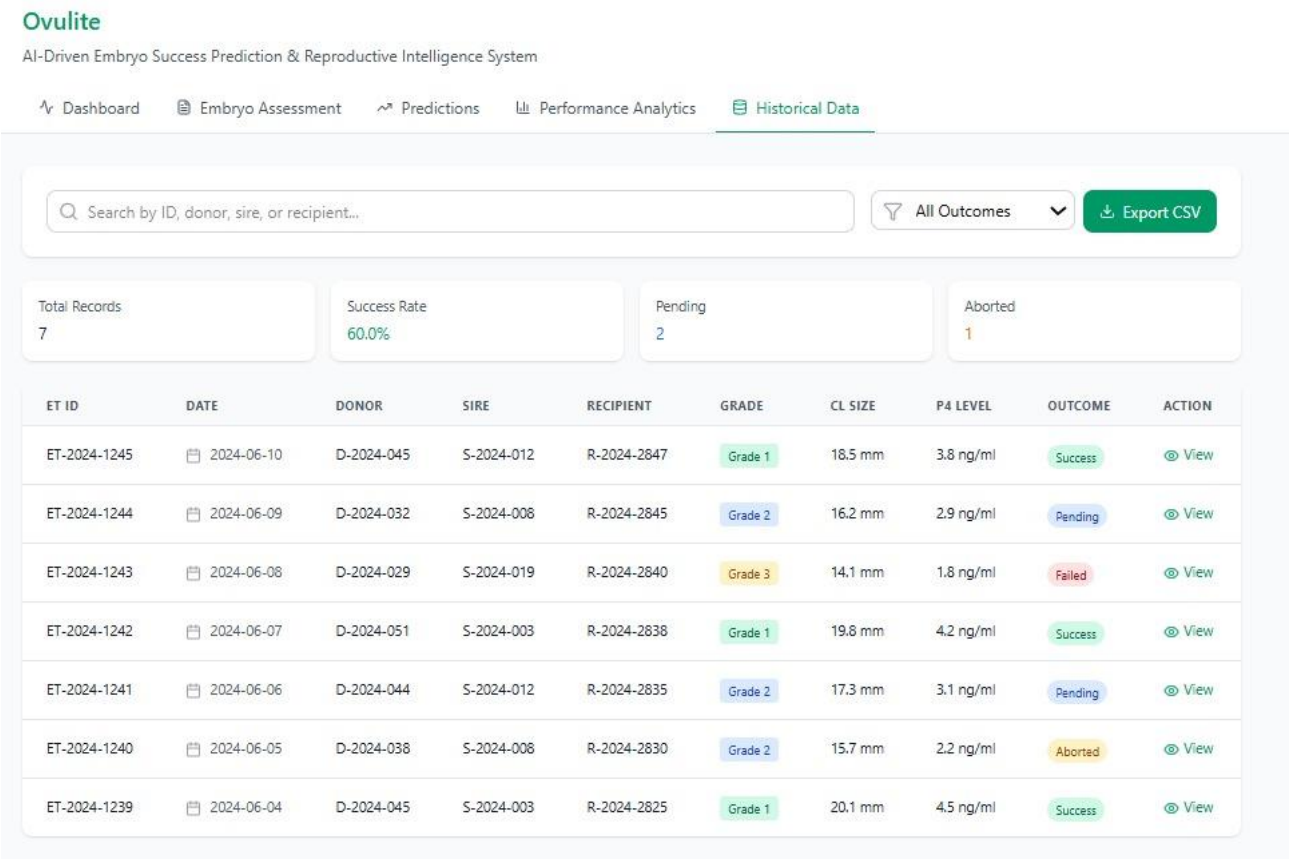
14. Mockups:


Ovulite
AI-Driven Embryo Success Prediction & Reproductive Intelligence System
Dashboard
Embryo Assessment
Predictions
Performance Analytics
Historical Data
Success Rate
68.5%
+5.2%
Active Pregnancies
142
+12
Abortion Risk Cases
8
-3
Embryos Assessed
1,247
+89
Monthly Success Rate Trend
80
60
40
20
0
Jan
Feb
Mar
Apr
May
Jun
Failure %
Success %
Embryo Grade Distribution
Grade 1: 342
Grade 2: 456
Grade 4: 160
Grade 3: 289
P4 Levels vs Success Rate
240
180
120
60
0
<1 ng/ml
1-2 ng/ml
2-4 ng/ml
>4 ng/ml
Successful
Total Cases
Recent Alerts & Recommendations
High Abortion Risk Detected
Recipient #2847 shows low P4 levels (1.2 ng/ml)
Technician Performance Improved
Dr. Ahmed's success rate increased to 74%
Media Batch Performance
Batch MB-2024-06 showing exceptional results


Ovulite
AI-Driven Embryo Success Prediction & Reproductive Intelligence System
Dashboard
Embryo Assessment
Predictions
Performance Analytics
Historical Data
New Embryo Assessment
Embryo Information
Donor ID
e.g., D-2024-001
Sire ID
e.g., S-2024-045
Embryo Day
Day 7
Embryo Grade
Select grade
Morphology Notes
Expanded blastocyst, clear ICM, etc.
Recipient Information
Recipient ID
e.g., R-2024-2847
CL Size (mm)
e.g., 18.5
Blood Flow Score (1-5)
Select score
P4 Level (ng/ml)
e.g., 3.2
Procedure Information
Technician ID
e.g., T-001
Media Batch ID
e.g., MB-2024-06
Additional Notes
Any additional observations or comments...
Assess & Predict
AI Prediction Results
Fill out the assessment form to receive AI-powered predictions


Ovulite
AI-Driven Embryo Success Prediction & Reproductive Intelligence System
Dashboard
Embryo Assessment
Predictions
Performance Analytics
Historical Data
AI Model Accuracy
Based on 4 confirmed outcomes
100.0%
4 / 4 accurate predictions
Search by ID, donor, or recipient...
All Status
ID
DATE
DONOR
RECIPIENT
GRADE
SUCCESS RATE
ABORTION RISK
STATUS
P-2024-001
2024-06-10
D-2024-045
R-2024-2847
Grade 1
82.5%
12.3%
Success
P-2024-002
2024-06-09
D-2024-032
R-2024-2845
Grade 2
68.2%
24.1%
Pending
P-2024-003
2024-06-08
D-2024-029
R-2024-2840
Grade 3
45.8%
38.5%
Failed
P-2024-004
2024-06-07
D-2024-051
R-2024-2838
Grade 1
87.1%
9.2%
Success
P-2024-005
2024-06-06
D-2024-044
R-2024-2835
Grade 2
72.4%
18.7%
Pending
P-2024-006
2024-06-05
D-2024-038
R-2024-2830
Grade 2
65.3%
28.2%
Aborted


Ovulite
AI-Driven Embryo Success Prediction & Reproductive Intelligence System
Dashboard
Embryo Assessment
Predictions
Performance Analytics
Historical Data
Donors
Sires
Technicians
Media Batches
Top Performing Donor
Simmental #051
Success Rate
80.0%
Donor Performance Comparison
32
24
16
8
0
Holstein #045
Jersey #032
Holstein #029
Simmental #051
Holstein #044
Successful
Total Transfers
Overall Success Rate Trend
80
60
40
20
0
Jan
Feb
Mar
Apr
May
Jun
Success Rate %
Detailed Donor Performance Statistics
ID
NAME
TOTAL TRANSFERS
SUCCESSFUL
SUCCESS RATE
PERFORMANCE
D-2024-045
Holstein #045
24
19
79.2%
Excellent
D-2024-032
Jersey #032
18
13
72.2%
Good
D-2024-029
Holstein #029
31
21
67.7%
Good
D-2024-051
Simmental #051
15
12
80.0%
Excellent
D-2024-044
Holstein #044
22
15
68.2%
Good



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(Directly supports CL size, blood flow, and pregnancy prediction modules.)

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